

Day 14 — week 2 review

MD Mutasim Billah · 52-week Data Science to ML/AI roadmap · no new script today, pure consolidation before Week 3

1. Why a review day, and why now

Days 8 through 13 introduced six days of new material back to back: cleaning, feature engineering, visualization, statistics, two days of SQL, and a full capstone project. That's a lot to actually retain. Today has no new syntax and no new script — the entire goal is finding out what didn't stick, while it's still cheap to fix. Week 3 builds directly on SQL and statistics, so gaps here turn into slow-downs there if they're not caught now.

2. Recap: Days 8-9 — cleaning and visualization

- Three different missing-value strategies for three different columns: median for Age (robust to outliers), drop for Cabin (too many missing to impute meaningfully), mode for Embarked (categorical, only 2 missing values)
- Four engineered features: FamilySize (SibSp + Parch + 1), IsAlone, AgeGroup (binned), Title (extracted from Name via regex)
- Six chart types practiced: bar, histogram, boxplot, scatter with trend line, and a 2x2 subplot dashboard combining several of these

3. Recap: Day 10 — statistics fundamentals

Try answering these without opening the script:

- What's the difference between population variance and sample variance, and why does sample variance divide by $n-1$ instead of n ?
- What do the IQR outlier fences actually compute? ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$)
- What does a p-value actually mean — and what's the most common misreading of it?
- What's the difference between a Type I error and a Type II error?

4. Recap: Days 11-12 — SQL, basic to intermediate

- WHERE filters individual rows, before any grouping happens. HAVING filters groups, after aggregation — that's why HAVING can reference COUNT(*) or AVG(...) and WHERE can't
- A CTE (WITH name AS (...)) is just a named subquery — same result as nesting a subquery in FROM, but readable top to bottom
- Window functions (RANK, ROW_NUMBER with OVER/PARTITION BY) keep every row and attach a rank to each one. GROUP BY collapses rows down into one row per group — very different behavior for a similar-looking task

- INNER JOIN drops rows that don't have a match in the other table. LEFT JOIN keeps every row from the left table regardless, filling unmatched columns with NULL

5. Recap: Day 13 — the full EDA portfolio project

The lesson of Day 13 was less about any single new function and more about pipeline order: clean once, run statistics on the clean data, cross-check those same findings in SQL, build the dashboard from the exact correlation matrix the statistics step already computed, then generate the README from the same finding sentences — never re-deriving anything, never touching raw uncleaned data again after step one.

6. Self-check

Answer these from memory, then go check:

- Why did Fare get a log transform in the dashboard, but Age didn't?
- Why use Welch's t-test (`equal_var=False`) instead of the textbook default?
- Write, from memory, a query that finds the top 2 highest fares per Embarked port using a window function
- Why was the README text generated by code instead of typed by hand after the fact?

7. Triage: what to do about gaps

If the self-check exposed a weak spot, the fix is targeted, not total. Shaky on one statistical test? Reread just that section of the Day 10 material, don't redo the whole day. Shaky on SQL? Rerun `day12_sql_intermediate.py` and change one value in each query to see what output shifts and why. Patch the specific gap — re-doing six days of work to fix one misunderstanding wastes far more time than it saves.

8. What's next

No script and no commit for Day 14 — review days don't produce new deliverables. Week 3 opens with Day 15: the first machine learning model, built on top of the exact same Titanic dataset this whole review just covered.